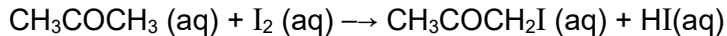


- 14** Propanone reacts with iodine in the presence of sulfuric acid.



The rate equation for this reaction is: $\text{rate} = k[\text{H}^+][\text{CH}_3\text{COCH}_3]$.

Two experiments were carried out. In both experiments, the initial concentrations of propanone and iodine remained the same but the initial concentration of the sulfuric acid was changed. The initial rate in the first experiment was three times faster than the initial rate in the second experiment. In the first experiment the initial pH was 1.50.

What is the initial pH in the second experiment?

- A** 1.02 **B** 1.98 **C** 2.28 **D** 4.50